ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

1.INTRODUCTION

• **Project Title:** Credit Card Approval System

• **Team Members:**

Pamulapati Chandana Sri (Team Lead)- ML model development and project integration.

Kurapati Hannah (Team Member)- Frontend.

Pathan Neelofer (Team Member)- Backend.

Konamanchili Nandu Priya (Team Member)- Testing, documentation, and presentation.

1.1 Project Overview

The **Credit Card Approval Prediction System** is a web application that uses machine learning to predict whether a credit card application will be approved or rejected. It analyzes applicant details such as income, employment, education, and other information to make the prediction.

The project is developed using **Python, Flask, HTML, CSS, JavaScript,** and **Machine Learning**. Users can enter their details through the web application and get the prediction result instantly.

This system makes the credit card approval process faster, easier, and more accurate.

1.2 Purpose

The main purpose of this project is to develop a machine learning-based system for predicting credit card approval.

Objectives:

- To automate the credit card approval process.
- To predict whether an application is approved or rejected.
- To reduce manual work and errors.
- To provide a simple and user-friendly application.
- To improve the speed and accuracy of the approval process.

This project shows how machine learning can help banks make faster and better credit card approval decisions.

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2. IDEATION PHASE

2.1 Problem Statement

Many banks still spend time reviewing credit card applications manually. This process can be slow and may lead to errors. There is a need for a system that can quickly analyze applicant details and predict whether a credit card should be approved or rejected.

2.2 Empathy Map Canvas

User: Bank Employee / Credit Card Applicant

Says:

- I want a quick approval process.
- The results should be accurate.

Thinks:

- Will my application be approved?
- The process should be fair and reliable.

Does:

- Enters the required details.
- Submits the application.
- Checks the prediction result.

Feels:

- Wants a fast response.
- Expects a simple and easy-to-use system.

2.3 Brainstorming

Possible ideas considered for the project:

- Manual approval system
- Rule-based approval system
- Machine learning-based prediction system

Among these, the machine learning approach was selected because it provides faster and more accurate predictions using historical data.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User opens the application.
2. User enters the required details.
3. The system processes the information.
4. The machine learning model predicts the result.
5. The prediction (Approved/Rejected) is displayed to the user.

3.2 Solution Requirement

Functional Requirements

- Accept applicant details from the user.
- Predict credit card approval.
- Display the prediction result.

Non-Functional Requirements

- Easy to use.
- Fast response.
- Accurate predictions.
- Secure user data.

3.3 Data Flow Diagram

Input → Data Processing → Machine Learning Model → Prediction Result

3.4 Technology Stack

Technology	Purpose
Python	Backend programming
Flask	Web framework
HTML	Web page structure
CSS	Styling

Technology	Purpose
JavaScript	Frontend interactions
Pandas & NumPy	Data processing
Scikit-learn / XGBoost	Machine learning model
MySQL	Database

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project uses a machine learning model to predict whether a credit card application should be approved or rejected. This reduces manual work and helps provide faster and more accurate results.

4.2 Proposed Solution

The proposed solution is a web application where users enter their details. The machine learning model processes the information and displays the prediction result on the screen.

4.3 Solution Architecture

User → Web Application → Machine Learning Model → Prediction Result

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was completed in the following stages:

Phase	Activity
Planning	Identified the problem and objectives
Data Collection	Collected and prepared the dataset
Model Development	Trained the machine learning model
Web Development	Developed the Flask web application
Testing	Tested the system and verified the results
Deployment	Finalized the project and generated predictions

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The system was tested using different applicant details to check its performance. The machine learning model correctly predicted whether the application was approved or rejected.

The application provides:

- Quick prediction results.
- Good accuracy.
- Easy-to-use interface.
- Smooth performance.

7. RESULTS

7.1 Output Screenshots

https://drive.google.com/file/d/1HYsP_Vyi7w8SEPOsfhy6QIlutjcYT41W/view?usp=sharing

(above is the link to show the codes and demo of both frontend and backend applications and there is demo of output also)

8. ADVANTAGES & DISADVANTAGES

Advantages

- Fast prediction process.
- Reduces manual work.
- Easy to use.
- Provides accurate results.

Disadvantages

- Accuracy depends on the dataset.
- The model may need updates with new data.
- Human verification may still be required.

9. CONCLUSION

- The Credit Card Approval Prediction System uses machine learning to predict whether a credit card application should be approved or rejected. The system provides fast and accurate predictions through a simple web application. It helps reduce manual work and supports better decision-making in the credit approval process.

10. FUTURE SCOPE

- Improve the model using larger and updated datasets.
- Add user login and authentication.
- Deploy the application on a cloud platform.
- Integrate the system with banking applications for real-time use.

11. APPENDIX

- **Source Code**
  The complete source code of the project is included with the project files.
- **Dataset Link**
  Kaggle Credit Card Approval Dataset:
  https://www.kaggle.com/datasets/rikdifos/credit-card-approval-prediction
- **GitHub Link**
  https://github.com/chandanasri-17/Credit-Card-Approval-Prediction
- **Project Demo Link**
  https://drive.google.com/file/d/1HYsP_Vyi7w8SEPOsfhy6QIlutjcYT41W/view?usp=sharing